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The effect of string diameter on damping of pendulums

Introduction

Pendulums are common instruments in our modern world, from entertaining toddlers on swings to keeping time to providing evidence for a round Earth. They are also one of the most common instruments used for experiments in school physics. What struck me was the variation in pendulums: we are taught that pendulums' periods do not change and that a pendulum should keep going forever, but as the first real life example we are given a swing, something we all know needs constant pushing to keep going. Looking into this led me to damping, and as I did further research on this phenomenon I came across a study on how the diameter of a string affects the damping constant of its pendulum[1]. However, aside from this study by Mohazzabi and Shankar from 2017, there is very little information or data on this phenomenon. Therefore, when given the opportunity to conduct an individual experiment of my choice, I decided I wanted to verify this study myself with a slightly altered method to be possible at school.

Research Question: How does the diameter of a wire affect the damping constant of the pendulum?

Background

Simple pendulum

A pendulum is an object that swings back and forth from a fixed point due to gravity. A simple pendulum is a pendulum consisting of a pendulum bob which can be treated as a point mass, connected to a string or rod of negligible mass swinging from a fixed point of suspension (Figure 1[2]).[3] A simple pendulum is a typical example of simple harmonic motion (SHM). This means that the pendulum has a restoring force acting on it, with a magnitude proportional to the displacement and always in the direction towards a fixed point. This fixed point is the vertical position where the bob rests when there is no displacement, and is therefore the point where displacement = 0. This is also known as the equilibrium position, as shown on Figure 1.

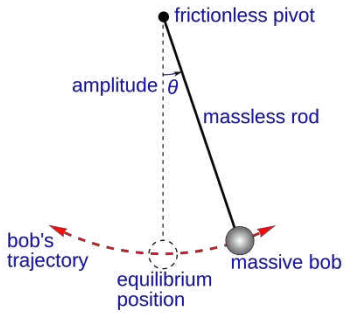


Figure 1

Damping

Although a simple pendulum should theoretically remain in motion indefinitely due to conservation of energy[4], in practice there are factors which prevent this, even if the pendulum used is relatively close to a perfect simple pendulum. This phenomenon is called damping. Damping occurs when the amplitude of an oscillation decreases due to dissipative forces. This means the energy of the system also decreases, and can be decreased until the system, the pendulum in this case, stops at a standstill.[5]

There are several types of damping. This investigation uses light damping. In light damping, oscillations still occur although the amplitude decreases with time. The decrease in amplitude can be modelled exponentially, as shown in Figure 2[6]. Although the amplitude decreases, the period of the pendulum stays constant.

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[1] Mohazzabi and Shankar, 2017
[2] Chetvorno, 2008
[3] Ling, Sanny and Moebs, 2019
[4] Woodford, 2018
[5] Hosch, n.d.
[6] Underdamped oscillations within an exponential decay envelope, n.d.

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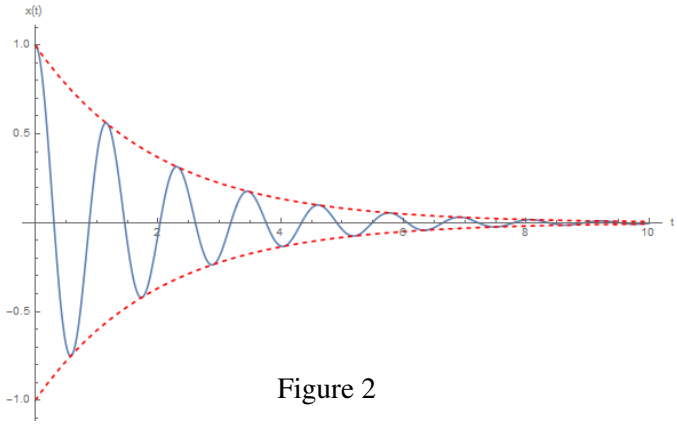


Figure 2

There are several dissipative forces which can cause damping. This investigation will focus on drag, also known as air resistance. This occurs when there is friction between the pendulum and air particles, which causes a loss of energy, primarily in the form of heat energy. Air resistance increases with size and surface area, and so generally the bob of the pendulum is seen as the main contributor to air resistance, while the string's effect is thought to be negligible.[7]

Derivation of the formula

There is no simple formula which connects the data that can be collected for this experiment given the available equipment. Therefore, this derivation is necessary to be able to analyse the data.

One formula for angular velocity is:

$$\omega = \frac{v}{r} = \frac{v}{l} \quad (1)$$

as in this case, the radius in the total length of the pendulum.

where ω is angular velocity (s^{-1}), v is linear velocity (m s^{-1}), r is the radius (m) and l is the length of the pendulum (m)

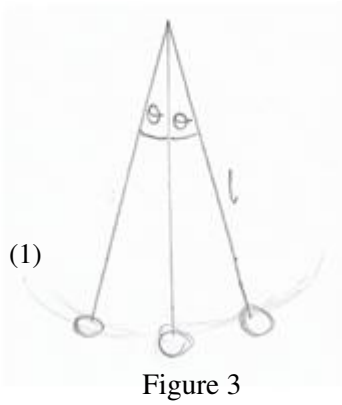


Figure 3

Angular velocity can also be calculated using a different formula. In the case of a pendulum, the position angle is 4 times the as θ shown on the diagram (Figure 3), as a whole oscillation consists of the pendulum swinging to the same amplitude at the other side and back. Therefore:

$$\omega = \frac{\theta}{T} = \frac{4\theta}{T} \quad (2)$$

where θ is the position angle (rad), and T is the period of the pendulum (seconds)

As both Formula (1) and (2) represent angular velocity, they can be put together to give the following formula:

$$\frac{v}{l} = \frac{4\theta}{T} \text{ which can be rearranged to } \theta = \frac{vT}{4l} \quad (3)$$

As the period and the length stay constant, only linear velocity needs to be measured to be able to calculate the amplitude. Velocity can be recorded using a data logger.

Formula (4), the solution to a second order differential equation, was taken from *Damping of a Simple Pendulum Due to Drag on Its String* by Mohazzabi and Shankar, 2017.

[7] Mohazzabi and Shankar, 2017

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$$\theta=\theta_0 e^{-kt/2} \quad (4)$$

where θ_0 is the initial amplitude (rad), θ is the amplitude at any given time (rad), k is the damping constant, and t is time (seconds)

This can be rearranged to give:

$$\ln\left(\frac{\theta}{\theta_0}\right) = -k/2\ t \quad (5)$$

This final formula, Formula (5) will result in a linear graph if $\ln\left(\frac{\theta}{\theta_0}\right)$ is graphed against t, from which k, the damping constant, can be calculated by dividing the gradient by -½.

Design

Hypothesis

The velocity will decrease over time for every diameter of wire as damping takes place. As velocity decreases, this also indicates a decrease in amplitude. Also, the thicker the wire, the more and quicker the velocity will decrease, and therefore the greater the damping constant of the wire.

Variables

Independent variable - diameter of the wire used as string for the pendulum. The different diameters used were: 150µm, 250µm, 300µm, 350µm, 550µm, and 650µm. All diameters were measured with a micrometer with an error margin of ±25µm.

Dependent variable - the velocity of a pendulum at its equilibrium position measured consistently for 90 seconds. The velocity was measured using a data logger and light gate, with an error margin of ±0.000000001 ($\pm10^{-9}$).

Control variables - mass of mass hanger (pendulum) - 200.33 ±0.01g; length of wire - 100 ±0.05cm; length of mass hanger - 19.1 ±0.05cm; place from which the pendulum was released - 30 ±0.05cm to the right. This means the angle from which the pendulum was released was also kept constant; how length of string was measured (from the bottom of the clamp to the top of the hook on the hanger); person letting go of the pendulum (to reduce random human error)

Procedure

1. Set up the apparatus as in the diagram (Figure 4) using the wire with diameter 150µm, taking care that the wire is 100cm from the bottom of the clamp to the top of the hook on the hanger. Wind the excess wire around the back of the stand so it does not interfere with the experiment. Make sure that when the pendulum is at rest, it hangs between the light gate.
2. Displace the bottom of the pendulum 20cm to the right.
3. Start the program on the data logger and release the pendulum. Ensure that the pendulum swings at 90° to the light gate so that the pendulum swings through and does not collide with the light gate.
4. Allow the pendulum to swing freely for 90 seconds, at which point the program on the data logger can be stopped. Send the results collected to a device on which they can be analysed.
5. Repeat this procedure two more times for the same wire.

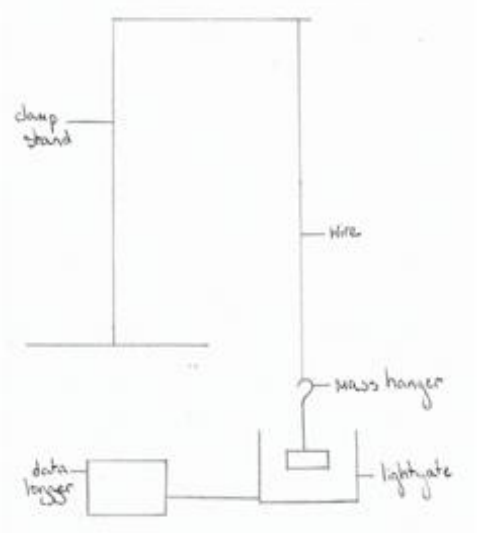


Figure 4

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6. Repeat steps 1-5 with wires of diameter 250µm, 300µm, 350µm, 550µm, and 650µm.

Risk Assessment

The safety issues in this experiment are minimal. The pendulum was swinging a little below head level as the experiment was conducted close to the ground due to the long string and nowhere higher to attach it. While the pendulum was swinging, a step backwards was taken to prevent any injury. Also, due to crouching it was hard to move away quickly, for example if something were to fall. The only object that might have fallen is the mass hanger, which was not likely to cause significant harm. There are no environmental or ethical issues with this method.

Results

Qualitative Data

For all wires, the pendulum swung back and forth consistently for 90 seconds before they were then stopped. For the thinner wires, a decrease in amplitude was not obvious from simple observation, although the thicker wires did show a visible decrease in amplitude. However, it is kept in mind that this might have been due to confirmation bias, and the damping effect could have been exaggerated or imagined. The quantitative data does, however, support the idea that damping occured.

Raw Data

For each wire, the velocity was measured once each oscillation for 90 seconds, which is the same as 41 oscillations. This was repeated for 3 trials for each wire. The table below shows an example of the raw data collected for Wire 1.

Run 1: Time (s) (±1E-5)	Run 1: Velocity (m/s) (±1E-9)	Run 2: Time (s) (±1E-5)	Run 2: Velocity (m/s) (±1E-9)	Run 3: Time (s) (±1E-5)	Run 3: Velocity (m/s) (±1E-9)
2.30852	0.341591303	2.31187	0.342676298	2.31113	0.352806390
4.49896	0.340730014	4.50260	0.346230023	4.50162	0.351738225
6.69023	0.339613350	6.69387	0.344769079	6.69284	0.350455616
8.88186	0.338782078	8.88469	0.345196467	8.88497	0.349630484
11.07250	0.337937735	11.07580	0.343777032	11.07622	0.348704084
13.26432	0.337606874	13.26714	0.342396543	13.26671	0.348596073
15.45464	0.336292709	15.45816	0.342168925	15.45753	0.347102039
17.64645	0.334999958	17.64962	0.341389399	17.64907	0.345839036
19.83732	0.334272081	19.83998	0.340597675	19.84083	0.345396373
22.02849	0.333216707	22.03192	0.338546065	22.03189	0.344559411
24.21963	0.332806390	24.22227	0.338674351	24.22227	0.343410035
26.41020	0.331771244	26.41331	0.338010457	26.41322	0.342927010
28.60199	0.331718968	28.60519	0.336419388	28.60482	0.342341489
30.79242	0.330496571	30.79539	0.335808437	30.79629	0.340862478
32.98425	0.328917614	32.98719	0.335571900	32.98709	0.341078825
35.17467	0.328374873	35.17760	0.334437789	35.17752	0.339414832
37.36566	0.327458188	37.36868	0.332972607	37.36855	0.338453589
39.55735	0.326965677	39.55989	0.332780620	39.56037	0.338053513
41.74769	0.326128609	41.75065	0.332836171	41.75159	0.336721301
43.93944	0.325764733	43.94209	0.331053129	43.94209	0.336681181
46.12959	0.324591015	46.13280	0.330218762	46.13261	0.335290816
48.32113	0.323297636	48.32420	0.328956957	48.32386	0.334062954
50.51212	0.322682134	50.51461	0.328878121	50.51558	0.333627097
52.70296	0.321802092	52.70608	0.327395055	52.70654	0.332614788
54.89418	0.321683046	54.89661	0.326684264	54.89673	0.331849501
57.08447	0.320479437	57.08761	0.326262246	57.08737	0.331298582
59.27631	0.319897633	59.27937	0.325221809	59.27878	0.330455360
61.46666	0.318981810	61.46927	0.324698199	61.47024	0.329298280
63.65829	0.317558609	63.66127	0.323301797	63.66101	0.329159627
65.84869	0.317168321	65.85136	0.323297044	65.85113	0.327869925
68.03936	0.316143055	68.04228	0.322115875	68.04198	0.327101727
70.23104	0.315691443	70.23370	0.321292785	70.23358	0.326762581
72.42112	0.314779694	72.42402	0.320838456	72.42482	0.325310883
74.61274	0.314233192	74.61557	0.320147844	74.61528	0.325287581
76.80281	0.313050284	76.80599	0.319205643	76.80552	0.323939820
78.99410	0.311866521	78.99707	0.317750488	78.99656	0.322837588
81.18523	0.311390671	81.18761	0.317771113	81.18827	0.322441459
83.37575	0.310614473	83.37874	0.316593166	83.37918	0.321372753
85.56708	0.310457770	85.56956	0.315745096	85.56932	0.320871776
87.75716	0.309209814	87.76035	0.315383740	87.75979	0.320108789
89.94882	0.308487256	89.95159	0.315196629	89.95101	0.319105674

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Processed Data

First, the average time and velocity per oscillation were calculated for each wire.

E.g. for the first oscillation for Wire 1:

Average time = (2.30852 + 2.31187 + 2.31113) / 3 = 2.31051 seconds to 5 decimal places

Error calculation: addition followed by division by constant, so addition of absolute error followed by multiplication by constant: (0.00001 + 0.00001 + 0.00001)×3 = ±0.00009

Average velocity = (0.341591303 + 0.342676298 + 0.352806390) / 3 = 0.345691330 m s^{-1} to 9 decimal places

Error calculation: addition followed by division by constant, so addition of absolute error followed by no change: (0.000000001 + 0.000000001 + 0.000000001)×3 = ±0.000000009

Average time (s) (±9E-5)	Average velocity (m/s) (±9E-9)	Amplitude (rad) (±0.08%)
2.31051	0.345691330	0.1677
4.50106	0.346232754	0.1679
6.69231	0.344946015	0.1673
8.88384	0.344536343	0.1671
11.07484	0.343472950	0.1666
13.26606	0.342866497	0.1663
15.45678	0.341854558	0.1658
17.64838	0.340742798	0.1653
19.83938	0.340088710	0.1649
22.03077	0.338774061	0.1643
24.22139	0.338296925	0.1641
26.41224	0.337569570	0.1637
28.60400	0.336826615	0.1634
30.79470	0.335722495	0.1628
32.98618	0.335189446	0.1626
35.17660	0.334075831	0.1620
37.36763	0.332961461	0.1615
39.55920	0.332599937	0.1613
41.74998	0.331895360	0.1610
43.94121	0.331166348	0.1606
46.13167	0.330033531	0.1601
48.32306	0.328772516	0.1595
50.51410	0.328395784	0.1593
52.70519	0.327270645	0.1587
54.89584	0.326738937	0.1585
57.08648	0.326013422	0.1581
59.27815	0.325191601	0.1577
61.46872	0.324326096	0.1573
63.66019	0.323340011	0.1568
65.85039	0.322778430	0.1565
68.04121	0.321786886	0.1561
70.23277	0.321248936	0.1558
72.42332	0.320309678	0.1553
74.61453	0.319889539	0.1551
76.80477	0.318731916	0.1546
78.99591	0.317484866	0.1540
81.18704	0.317201081	0.1538
83.37789	0.316193464	0.1534
85.56865	0.315691547	0.1531
87.75910	0.314900781	0.1527
89.95047	0.314263186	0.1524

The table shows the average time, velocity and amplitude for Wire 1

Using these values, the amplitude for each oscillation was calculated. For this, Furmula (3) was used. The value used for the period is the average time taken for the first oscillation to be completed, regardless of which oscillation the amplitude is being calculated for, as the period should stay constant. For length, the total length of the string and the mass hanger was used.

E.g. for the first oscillation for Wire 1:

$$\theta = \frac{vT}{4l} = \frac{0.345691330 \times 2.31051}{4 \times 1.191} = 0.1677 \text{ rad to 4 significant figures}$$

Error calculation: separate multiplication and multiplication by a constant followed by division, so separate addition of fractional error and no change in fractional error followed by addition of fractional error:

$$\frac{0.000000009}{0.345691330} + \frac{0.00009}{2.31051} = \pm 4 \times 10^{-5} \text{ to 1 significant figure}$$

$$\frac{0.0005+0.0005}{1.191} = \pm 0.0008 \text{ to 1 significant figure}$$

$4 \times 10^{-5} + 0.0008 = \pm 0.0008$ to 1 significant figure = 0.08%

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Next, the formula taken from the original study was applied so that the results could be graphed and a value for the damping constant shown. For this the original amplitude was needed. This was calculated trigonometrically using the length of the pendulum and the displacement.

$\sin\theta_0 = 0.20 / 1.191$

$\theta_0 = \sin^{-1}(0.200 / 1.191) = 0.169$ rad

Error calculation: division so addition of fractional error: $\frac{0.0005}{0.200}+\frac{0.0005}{1.191}$ = ±0.003 to 1 significant figure = ±0.3% = ±0.0005 rad to 1 significant figure

E.g. for the second oscillation for Wire 1:

$\ln(\frac{\theta}{\theta_0}) = \ln(\frac{0.169}{0.1679}) = -0.00641$ to 3 significant figures

Error calculation: division so addition of fractional error: 0.003 + 0.0008 = ±0.004 to 1 significant figure = ±0.4% = $\pm 3\times10^{-5}$ to 1 significant figure

As can be seen in the table, an average of all the times for all 6 wires was taken in order to graph all results on one graph later. It is acceptable to take an average as the period for each wire should have been the same, considering neither the length nor the gravity were changed.

	ln (θ / initial θ) (±0.4%)					
Average time (s) (±3E-3)	Wire 1	Wire 2	Wire 3	Wire 4	Wire 5	Wire 6
4.47463	-0.00641	-0.0103	-0.00917	-0.00903	-0.00608	-0.00633
6.65321	-0.0101	-0.0111	-0.0111	-0.00991	-0.00870	-0.0139
8.83197	-0.0113	-0.0151	-0.0166	-0.0132	-0.0145	-0.0165
11.01047	-0.0144	-0.0175	-0.0163	-0.0161	-0.0180	-0.0192
13.18892	-0.0162	-0.0199	-0.0207	-0.0198	-0.0211	-0.0222
15.36736	-0.0191	-0.0207	-0.0236	-0.0227	-0.0220	-0.0248
17.54606	-0.0224	-0.0235	-0.0255	-0.0256	-0.0256	-0.0284
19.72446	-0.0243	-0.0261	-0.0296	-0.0278	-0.0282	-0.0318
21.90294	-0.0282	-0.0293	-0.0304	-0.0317	-0.0322	-0.0335
24.08137	-0.0296	-0.0291	-0.0342	-0.0353	-0.0346	-0.0375
26.25984	-0.0317	-0.0321	-0.0385	-0.0383	-0.0371	-0.0409
28.43834	-0.0340	-0.0350	-0.0390	-0.0405	-0.0402	-0.0442
30.61671	-0.0372	-0.0382	-0.0426	-0.0427	-0.0429	-0.0461
32.79528	-0.0388	-0.0396	-0.0454	-0.0456	-0.0466	-0.0501
34.97359	-0.0422	-0.0410	-0.0477	-0.0488	-0.0493	-0.0536
37.15199	-0.0455	-0.0446	-0.0524	-0.0515	-0.0524	-0.0562
39.33050	-0.0466	-0.0475	-0.0531	-0.0541	-0.0552	-0.0590
41.50894	-0.0487	-0.0498	-0.0572	-0.0570	-0.0582	-0.0626
43.68734	-0.0509	-0.0510	-0.0600	-0.0603	-0.0609	-0.0664
45.86560	-0.0543	-0.0544	-0.0624	-0.0636	-0.0644	-0.0689
48.04409	-0.0582	-0.0573	-0.0661	-0.0662	-0.0670	-0.0713
50.22254	-0.0593	-0.0595	-0.0677	-0.0685	-0.0704	-0.0753
52.40086	-0.0627	-0.0610	-0.0712	-0.0711	-0.0730	-0.0794
54.57912	-0.0644	-0.0642	-0.0752	-0.0741	-0.0762	-0.0817
56.75751	-0.0666	-0.0669	-0.0764	-0.0770	-0.0795	-0.0839
58.93607	-0.0691	-0.0705	-0.0806	-0.0796	-0.0823	-0.0885
61.11426	-0.0718	-0.0711	-0.0834	-0.0823	-0.0856	-0.0925
63.29263	-0.0748	-0.0737	-0.0863	-0.0855	-0.0887	-0.0948
65.47096	-0.0766	-0.0773	-0.0901	-0.0887	-0.0918	-0.0973
67.64935	-0.0796	-0.0804	-0.0918	-0.0916	-0.0952	-0.102
69.82768	-0.0813	-0.0817	-0.0960	-0.0942	-0.0981	-0.106
72.00585	-0.0842	-0.0847	-0.0993	-0.0968	-0.101	-0.108
74.18435	-0.0855	-0.0880	-0.102	-0.0997	-0.105	-0.111
76.36264	-0.0892	-0.0916	-0.106	-0.103	-0.108	-0.115
78.54088	-0.0931	-0.0937	-0.108	-0.105	-0.111	-0.120
80.71911	-0.0940	-0.0957	-0.112	-0.108	-0.114	-0.122
82.89756	-0.0972	-0.0992	-0.115	-0.111	-0.118	-0.125
85.07592	-0.0988	-0.102	-0.118	-0.114	-0.121	-0.129
87.25416	-0.101	-0.106	-0.120	-0.118	-0.124	-0.134
89.43269	-0.103	-0.106	-0.123	-0.120	-0.128	-0.136

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E.g. for the second oscillation

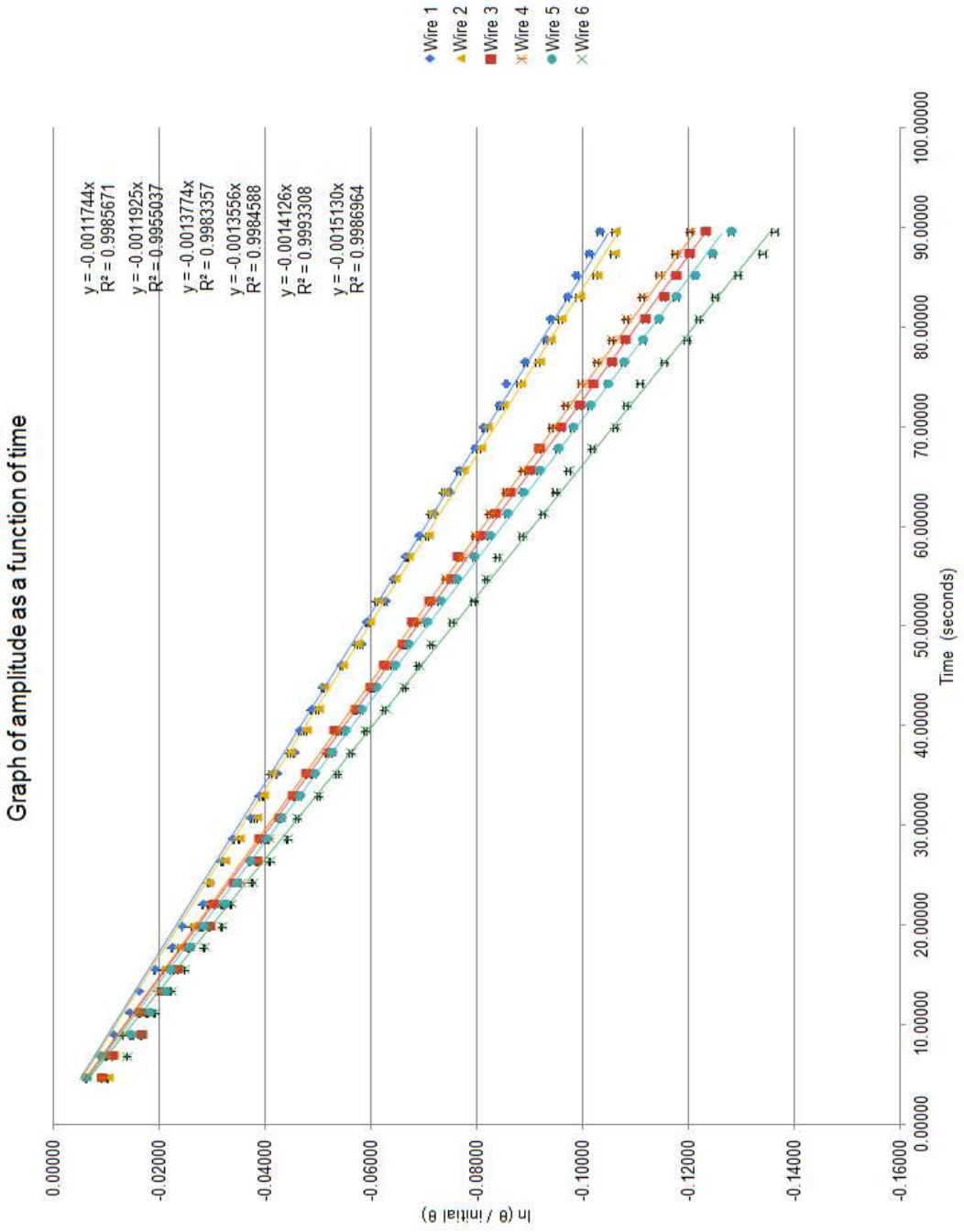
(4.50106 + 4.49705 + 4.45381 + 4.49362 + 4.42095 + 4.48127) / 6 = 4.47463 to 5 decimal places

Error calculation: addition followed by division by constant, so addition of absolute error followed by multiplication by constant: (0.00009 + 0.00009 + 0.00009 + 0.00009 + 0.00009 + 0.00009)×6 = ±0.003 to 1 significant figure

Also, the first oscillation is excluded as there should not have been any damping at the beginning, so the amplitude should have been 0.169. The logarithm of 1 (or 0.169 / 0.169) is zero, and so instead of including the first oscillation, the lines of best fit on the graph were simply set to go through 0.

The lines of best fit are in order from the smallest wire at the top to the biggest wire at the bottom.

Note that error bars were drawn in both directions, although they are often so small that they cannot easily be seen, particularly the horizontal ones.



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To find the damping constant, the gradient must be multiplied by -2.

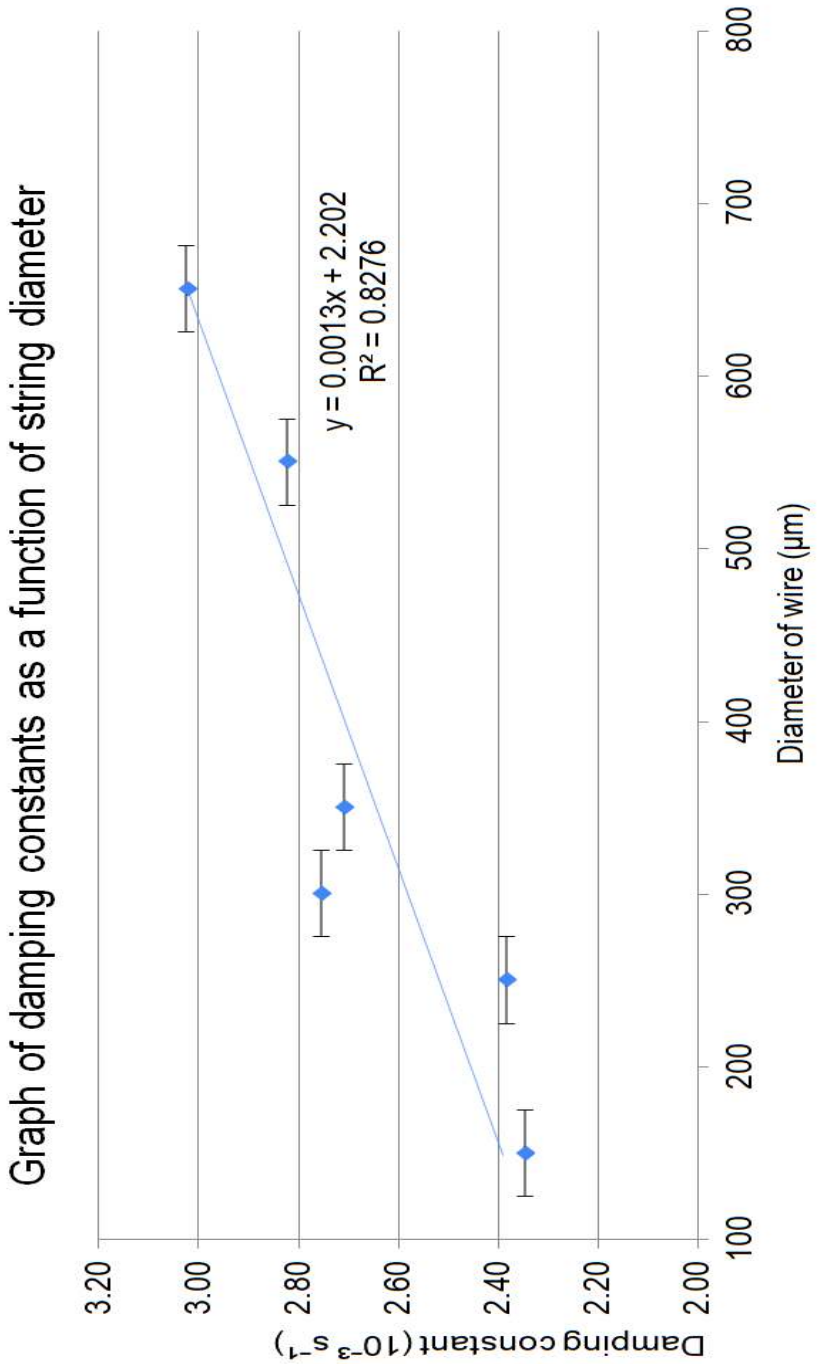
E.g. for Wire 1: -0.00117×-2 = 0.00235 to 3 significant figures

Diameter of wire (μm) (±25)	Damping constant (10^{-3} s^{-1})
150	2.35 ±0.0000046
250	2.39 ±0.0000036
300	2.75 ±0.0000055
350	2.71 ±0.0000054
550	2.83 ±0.0000057
650	3.03 ±0.0000061

Error calculation: the error of a gradient is half the difference between the minimum and maximum gradient. These minimum and maximum gradients were drawn on a separate graph for each wire to avoid clutter on the above graph. There are several issues with this method of error calculation that will be discussed in the evaluation.

E.g. for Wire 1: $\frac{maximum\ gradient - minimum\ gradient}{2} = \frac{(-0.001151)-(-0.0011602)}{2} = \pm 0.0000046$ to 2 significant figures

The damping constant was then graphed against the diameter of the wire. Vertical error bars are present but are so small they are barely visible.



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Discussion

Although the distribution of points does not demonstrate a perfect straight line, the correlation coefficient is 0.8276, which is high enough to establish that the relationship shown by the graph is accurate to a degree where conclusions can be drawn from it. The graph shows a linear relationship and suggests that the damping constant is directly proportional to the diameter of a wire, the damping constant increasing as the diameter of the wire increases.

One of the contributors to damping and therefore the damping constant is air resistance. Air resistance is a force that opposes the motion of an object through the air, and can be explained as particles in the air colliding with the moving object.[8] It therefore makes sense that a bigger surface area allows for more air particles to collide with it which therefore cause a bigger opposing force, leading to greater damping. This is also shown in the formula:

$$F_D = \frac{1}{2} \rho v^2 C_D A \quad (6)$$

where F_D is the drag force (air resistance), ρ is density of the fluid, v is the speed of the object, C_D is the drag coefficient, and A is the cross-section area of the object[9]

As the experiment uses air as fluid and was done in a constant location, the density can be assumed to be constant. The initial speed is also constant as the pendulum always had the same mass and initial amplitude. The drag coefficient depends on the shape of the object, which was also constant. Therefore, the cross-section area is the only thing affecting air resistance in this experiment.

The cross-section area of a wire would be its length multiplied by the diameter, resulting in a long, thin rectangle. Therefore, surface area increases proportionally to the diameter, meaning air resistance does as well. At 650µm, the thickest wire is more than 4 times thicker than Wire 1 (150µ), and so there is an increase by the same factor in its area and air resistance, making it understandable that damping changes

As this investigation is based on Mohazzabi and Shankar's *Damping of a Simple Pendulum Due to Drag on Its String*, the results can be compared. The results from Mohazzabi and Shankar's study are given below.

Diameter of wire (µm)	Damping constant (10^{-3} s^{-1})
250	0.902
279	0.937
428	1.075
570	1.195
718	1.256

The damping constants calculated in this investigation are a lot higher than in the original study. For example, Mohazzabi and Shankar calculate a damping constant for a wire with diameter 250µm of 0.902, while this experiment gets 2.39. This is particularly surprising as the original study used a wire twice the length of this investigation, and so surface area is twice as large as well. This trend of getting results of more than double what the original study got when they should be half stands for all diameters. This can be explained by the fact that these damping constants take into account the bob as well as the string. The original study uses a small sphere as bob, while here a mass hanger was used, which might have contributed significantly to the damping constant. Interestingly, the original study notes 0.367 as its average speed, which is higher than any initial speed here. The higher the speed, the greater the air resistance and therefore damping should be. This suggests that there were factors at work during this experiment that have not yet been considered which may have affected the results. These will be discussed in the evaluation.

[8] Williams, 2016
[9] Williams, 2016

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Evaluation

The biggest issue with the results is, as previously mentioned, their magnitude compared to the only other available values for this topic. Not only are they bigger, which might be written off as the contribution of the bob, but the rate at which they increase is also greater. In this investigation, a 500µm increase in diameter (from 150 to 650µm) resulted in an increase of 0.68×10^{-3} s^{-1} in the damping constant, while a similar increase in the original investigation (from 250 to 718µm) only resulted in a 0.35×10^{-3} s^{-1} increase in the damping constant. This means the increase here is almost twice that of the original study.

The errors in this experiment are small thanks to the precision of equipment, and therefore do not cover the deviation from the expected results. The most likely explanation is the friction at the pivot point, another large source of damping. First of all, as the wires were attached to the clamp directly by wrapping it around the clamp, this causes friction, and the bigger the surface area, the more friction. This would cause an increase in damping as the diameter increases that is additional to the change caused by air resistance. Another explanation is the fact that the thicker wires could not be tied as effectively and so were often wrapped around the clamp a few more times to make sure they were secure. This significantly increased surface area of the wire in contact with the clamp, and therefore friction. This is a problem that Mohazzabi and Shankar most likely avoided using better equipment and nylon string, rather than wires. A problem like this could be reduced by using a split cork to hold the end of the wire/string, so each wire is secured the same way.

As mentioned in the analysis, there was a small issue with the error which stems from having to acquire an error for the gradient of a graph. The problem is that for some of the wires, the gradient based on the experimental data does not actually fall between the theoretical minimum and maximum gradient. This is because the minimum and maximum are solely based on the first and last data points whereas the line of best fit includes all data points. Therefore, if either the first or last data point is off, it has a much bigger effect on the maximum and minimum slopes than on the line of best fit. Also, even if the gradient does fall between the minimum and maximum, it most likely does not fall perfectly in the middle, and so ±half-of-the-difference is an approximation. However, this approximation was used to make error calculations possible for all wires. The errors are of such a small magnitude that this approximation does not have a big impact anyway.

Aside from this, a few additional issues were noticed while conducting the experiment, summarised in the table below. These might have contributed to the difference in this experiment's results and the original study, or to the imperfect coefficient of determination.

Problem	**Correction**	**Effect on results**	**Type of error**
The wires, particularly the thicker ones, had nicks in them.	Ideally, real string would be used, such as the nylon in the original study. Unfortunately, no strings of the same material but different diameter were available.	If the nicks were present before measuring the wire, the wire would have had a larger surface area and increased drag. If nicks appeared later, the wire would be under 1m and so some calculations, such as those of initial angle, would be off, effecting all subsequent calculations done for this wire as well.	random
As the wires were coiled, while measuring them it	Again, using normal string would work	If the wires were a little coiled rather than straight, they would have been a	random

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could be hard to get them straight, and so errors could have been made in the measurements.	better, as they are not likely to have a shape. Also, using two people for the job of measuring may work better as someone can hold the wire stretched out.	little over 1m. As displacement was kept the same, this would have resulted in a smaller initial amplitude than believed. As the initial amplitude was assumed to be 0.169, if it was actually less it would have seemed to drop quickly, making it seem like large damping. On the one hand, damping would decrease, as the velocity would be lower.	
The θ changes during one oscillation, in the sense that the pendulum will not reach as high on the other side of the pendulum as at the previous side. Damping is happening constantly, so the calculated value takes the average of θ for one oscillation.	It is not possible to solve an issue like this with the equipment available. This would require constant tracking of the motion of the pendulum. Certain motion sensors might be capable.	This does not necessarily affect the results, as the investigation looks at the damping over one oscillation and not throughout an oscillation. The investigation can still effectively show how the amplitude changes over time. However, it does not take into account damping over the course of one period, which might be interesting to look at.	systematic
As the pendulum had to be re-setup for each wire, the equilibrium position may not always have been perfectly aligned with the light gate, and therefore the point where velocity is measured.	Fixing both the clamp and the light gate in place would improve this a lot. Then, even if the wire was attached a little differently, this would not have much of an effect. As it was, the clamp, light gate and pendulum all had to be aligned for each experiment, resulting in a variation of setups.	The effect on the results is not big, but nonetheless present. Regardless of where the velocity is measured, as long as this is constant the change in velocity measured should be the same. However, it can be argued that if the velocity was measured at a different point in the oscillation, more or less of the damping could have taken place than at the equilibrium point. This could mean that the drop from the initial amplitude is either calculated as more or less significant than it should be, which can affect the perspective of the results.	random
The pendulum was not always displaced exactly 20cm.	The ruler measuring displacement could be attached at the level at which the bob should hang when it is released, not below, which requires some estimating of where exactly 20cm is.	The initial amplitude would not have been the same as the calculated value, leading the amplitude to seemingly drop fast (as the initial displacement was less), or take a long time to drop as it is actually dropping from higher than assumed (if displacement is greater than assumed). Also, the total energy of the system would be different, due to greater or smaller initial gravitational potential energy, affecting velocity, which is critical to this investigation.	random

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Conclusion

In conclusion, the bigger the diameter of a string, the greater its damping coefficient in a pendulum. However, damping depends on a lot of factors such as the shape of the bob, the length of the string and other forces such as friction acting on the pendulum.

Suggestions for further investigation

First of all, it is suggested that this experiment is carried out at least one more time, to eliminate some of the extraneous variables this investigation struggled with. This most notably includes the friction at the pivot.

It would be interesting to see how the different shapes of bobs affect damping, looking not just at cross-section area, but also factors such as straight or rounded edges (sphere in Mohazzabi and Shankar's study was completely rounded, while this experiment's bob had edges). Another interesting factor would be weight distribution of the bob.

Lastly, looking at the material of the string might be of interest. One of the most notable differences between this and the original study is the strings used, and it is wondered whether this may have had an effect on the difference in damping constants.

Bibliography

Chetvorno (2008). *Diagram of simple gravity pendulum*. [image] Available at: https://commons.wikimedia.org/wiki/File:Simple_gravity_pendulum.svg [Accessed 25 Sep. 2019].

Ling, S., Sanny, J. and Moebs, B. (2019). *15.4: Pendulums*. [online] Physics LibreTexts. Available at: https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_University_Physics_(OpenStax)/Map%3A_University_Physics_I_-_Mechanics%2C_Sound%2C_Oscillations%2C_and_Waves_(OpenStax)/15%3A_Oscillations/15.4%3A_Pendulums [Accessed 25 Sep. 2019].

Woodford, C. (2018). *How do pendulum clocks work?*. [online] Explain that Stuff. Available at: https://www.explainthatstuff.com/how-pendulum-clocks-work.html [Accessed 23 Oct. 2019].

Elsea, P. (1996). *Simple Harmonic Motion*. [online] Artsites.ucsc.edu. Available at: http://artsites.ucsc.edu/EMS/music/tech_background/TE-12/teces_12.html [Accessed 23 Oct. 2019].

Nave, R. (n.d.). *Pendulum*. [online] Hyperphysics.phy-astr.gsu.edu. Available at: http://hyperphysics.phy-astr.gsu.edu/hbase/pend.html [Accessed 25 Sep. 2019].

Courses.lumenlearning.com. (n.d.). *The Simple Pendulum | Physics*. [online] Available at: https://courses.lumenlearning.com/physics/chapter/16-4-the-simple-pendulum/ [Accessed 25 Sep. 2019].

Hosch, W. (n.d.). *Damping | physics*. [online] Encyclopedia Britannica. Available at: https://www.britannica.com/science/damping [Accessed 26 Sep. 2019].

Underdamped oscillations within an exponential decay envelope. (n.d.). [image] Available at: https://brilliant.org/wiki/damped-harmonic-oscillators/ [Accessed 26 Sep. 2019].

Mini Physics. (n.d.). *Damping*. [online] Available at: https://www.miniphysics.com/damping.html [Accessed 26 Sep. 2019].

Mohazzabi, P. and Shankar, S. (2017). Damping of a Simple Pendulum Due to Drag on Its String. *Journal of Applied Mathematics and Physics*, 05(01), pp.122-130.

Williams, M. (2016). What Is Air Resistance? - Universe Today. Retrieved 25 October 2019, from https://www.universetoday.com/73315/what-is-air-resistance/